

On the performance of machine learning architectures for RNA structure prediction

Bachelor thesis defense

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Overview

1. Research question intro

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2. RNA basics

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2. RNA basics
3. Models I reviewed

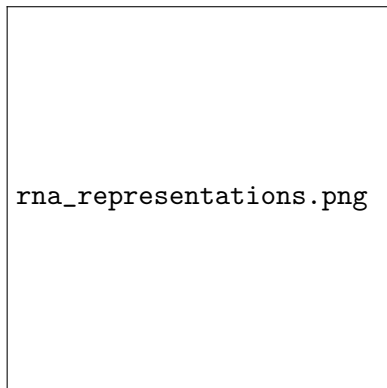
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3. Models I reviewed
4. Pipeline

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5. Results

The question behind the thesis



RNA structure as something we can
compute with

Many modern RNA folding tools report strong scores.

But those tools often combine neural networks with **constraints**, **preprocessing**, **postprocessing**, and **dataset effects**.

So what can the architecture itself learn?

Two research questions

1. **How much data is necessary** for FFN, CNN, and BLSTM models to perform sufficiently well?

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1. **How much data is necessary** for FFN, CNN, and BLSTM models to perform sufficiently well?
2. **How do these architectures behave on biased datasets** such as base-pair span, GC content, helix length, and multiloops?

⇒ **Not just “which model wins”, but what each architecture learns.**

What is RNA?



rna_ensemble.png

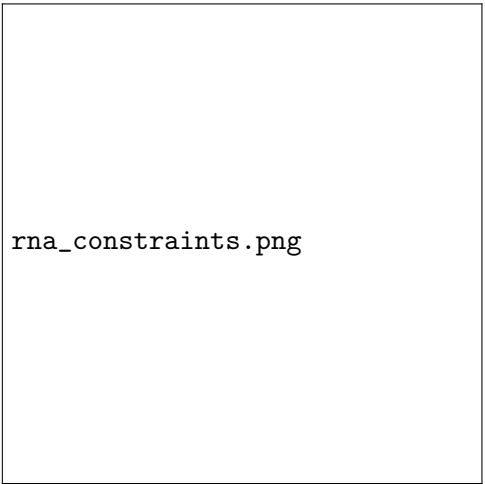
RNA is a sequence over

$$\{A, C, G, U\}$$

It folds by forming base pairs.

The relevant abstraction in this thesis is the **secondary structure**: which positions pair with which other positions.

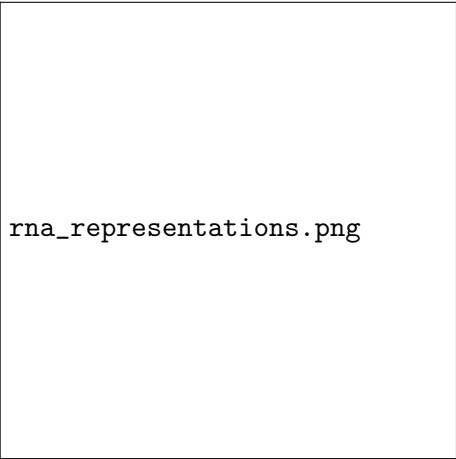
RNA secondary structure has rules



`rna_constraints.png`

canonical pairs, minimum loop length, one partner per base, no pseudoknots

From sequence to base-pair matrix



rna_representations.png

A base-pair matrix is an $L \times L$ image-like representation.

Cell (i, j) says how likely base i pairs with base j .

This makes RNA folding look like a sparse matrix prediction problem.

The baseline target: RNAfold / Zuker-style folding

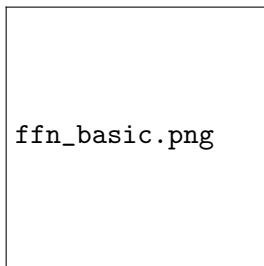
Classical folding: find a structure with low free energy.

$$p(s) = \frac{1}{Z} e^{-E(s)/(kT)}$$

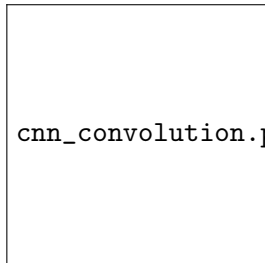
In this thesis, synthetic sequences are generated and their target structures are produced by **ViennaRNA RNAfold**.

So the networks are trained to approximate this biochemical model.

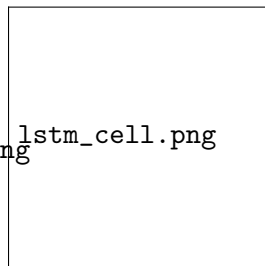
The three modeling principles



FFN
fully connected feature
integration

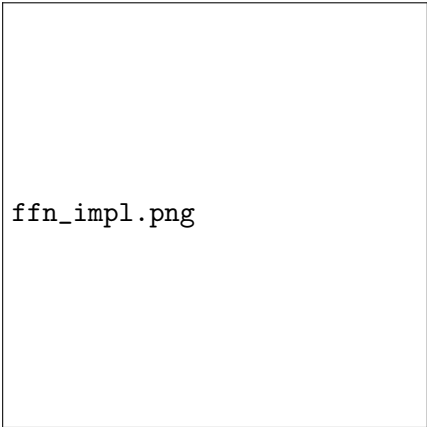


CNN
local pattern extraction



BLSTM
sequential dependency
modeling

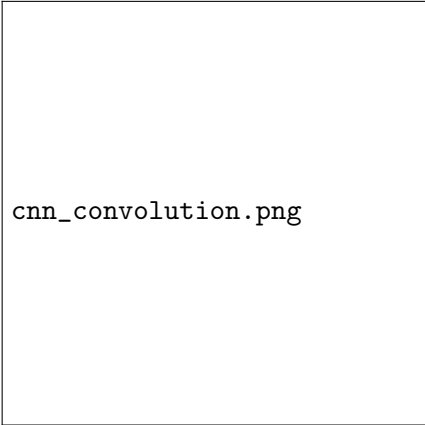
FFN: fully connected, but no structural prior



ffn_impl.png

one-hot encoded sequence $\rightarrow$ hidden layers $\rightarrow L^2$ output neurons

CNN: local kernels on sequence and pairwise space



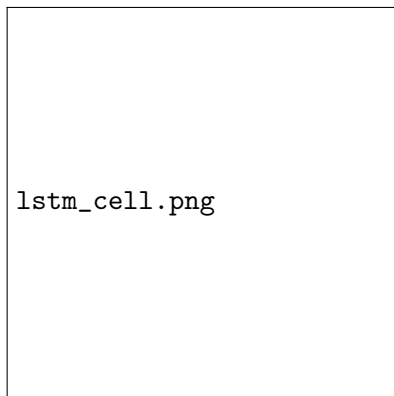
cnn_convolution.png

Kernels learn reusable local patterns.

For RNA this suggests a bias towards motifs like stems, hairpins, and local constraints.

High capacity: about 5 million trainable parameters in the optimized model.

BLSTM: sequence context in both directions



A BLSTM reads the RNA from both directions.

Each nucleotide representation can contain left and right sequence context.

The pair score is then computed from two position representations.

Design constraints I chose deliberately



synthetic data
controlled biases,
RNAfold labels

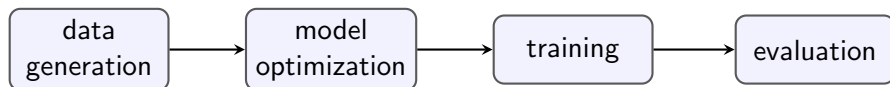
100

fixed length
fair FFN comparison, no
padding effects



no dropout / weight
decay
expose architecture
behavior

Pipeline: from data to evaluation



The framework was written in Python/PyTorch and keeps the experimental loop identical across architectures.

RQ1 setup: how much data is enough?

train size	200	1,000	5,000	10,000	50,000
idea	tiny	small	moderate	large	scaling

Each optimized architecture is trained on each size and evaluated on an independent 10,000-sequence test set.

Primary score: **MCC**. Additional RNA metrics: symmetry, sparsity, non-canonical ratio, minimum loop length.

RQ1 result: MCC scaling

Model	200	1,000	5,000	10,000	50,000
FFN	0.0104	0.0184	0.0257	0.0272	0.0280
CNN	0.2711	0.3165	0.3997	0.3976	0.4260
BLSTM	0.0000	0.0000	0.2924	0.3043	0.3359

RQ1 result: MCC scaling

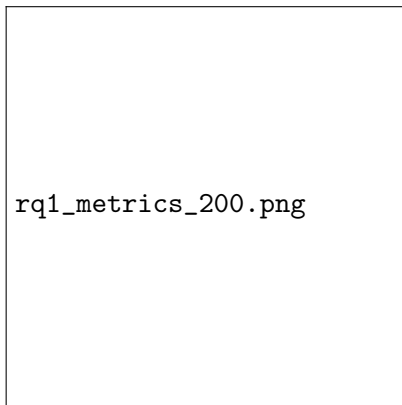
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⇒ **CNN works early; BLSTM needs roughly 5,000; FFN stays near zero.**

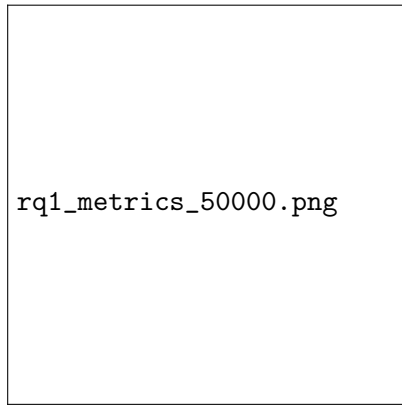
RQ1 visual result: what the models predict

rq1_bpm_table.png

RQ1 RNA metrics: low vs. high data



train size 200



train size 50,000

RQ1 interpretation by architecture

FFN	learned an average base-pair distribution, not a sequence-specific folding rule
CNN	strong with little data, good local constraints, but clear overfitting
BLSTM	no useful thresholded predictions for 200/1,000; starts around 5,000; weak on minimum loop length

⇒ **Architecture bias explains a lot of the behavior.**

RQ2 setup: biased datasets

Models were trained and evaluated across controlled biases:

`rq2_avg_structures.png`

- ▶ base-pair span
- ▶ GC content
- ▶ helix length
- ▶ multiloop topology

RQ2: base-pair span and GC content

rq2_bp_gc_tables.png

RQ2: helix length and multiloops

rq2_hl_multi_tables.png

RQ2 interpretation

FFN	mostly follows the average picture of train vs. evaluation set
BLSTM	more sensitive to base-pair span and GC-content shifts
CNN	more sensitive to helix-length bias, likely due to kernel-learned motifs
Multiloops	differences are moderate; MCC may hide topology errors

Overall take-home

Plain architectures alone do **not** reach engineered RNA folding tools.

But they reveal characteristic strengths and failure modes.

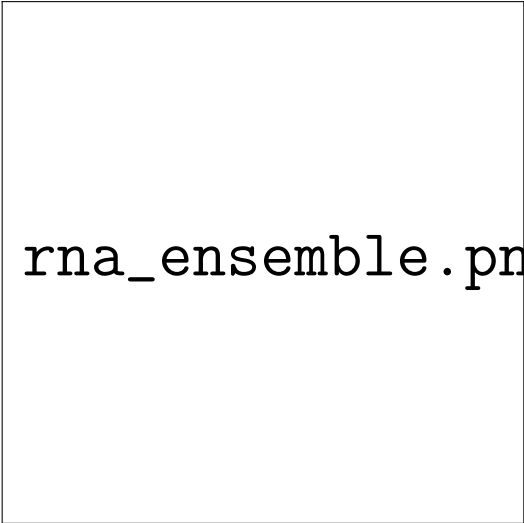
Architecture	main lesson
FFN	learns dataset average
CNN	good local motifs, overfits
BLSTM	needs more data, global context

Outlook

- ▶ hybrid architectures
- ▶ equal parameter-count comparison
- ▶ multiple random seeds
- ▶ finer bias sweeps
- ▶ real biological RNA datasets
- ▶ variable-length sequences
- ▶ transformers / graph neural networks
- ▶ topology-aware multiloop metrics

Danke für die Aufmerksamkeit!

Questions?



rna_ensemble.png